Email Backend: Conversation Log

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The chat between you and Claude about the email backend, in order, without the content of the plan and schema docs.

1. The plan and the mailbox pool

You: i need to make a backend for sending and receiving mail through my system, mail delivaried, bounce all info need to know and store on my db, i will get info like receipents a array, email body and send via business email of prime force or normal mail type from a mcp server. in my backend business main and normal mail are stored with those mail send mail randomly, its needs to handle 2k-5k mails daily, also if any receipant reply that also needs to store in my backend server and also need to trace for with repeipant which sender mail goes... now give me a plan for it in a md file.., also need a email provider costing details for this,

Claude: Created the Email Backend Plan doc (exportable as Markdown), covering architecture, sender rotation, sending pipeline, delivery and bounce tracking, reply ingestion, database schema, API, deliverability, provider costing, and a roadmap.

You: it not send 2000-5000 by multiple senders like round-robin or something. i have 30 business mail and 1500 normal email so plan for it

Claude: I've updated the plan for your 30 business and 1,500 normal mailboxes.

The Sender pool section now uses round-robin with a cooldown. Each mailbox goes to the back of its pool's line after it sends, and a Redis script makes sure no mailbox goes over its daily cap. At 5,000 emails a day, each business mailbox sends about 40 and each normal mailbox about 2 to 3. The section also has a 4-week warm-up schedule, a status flow for locked accounts, and how to run 1,530 logins.

I also updated the overview, the send request (an `overflow` setting for business-only requests), the schema, the API (bulk mailbox import), the costing, and the architecture diagram to match.

One risk to know: Gmail and Outlook.com don't allow automated bulk sending from free accounts, so expect some of the 1,500 to get locked over time. The plan tracks this and moves those recipients to a new sender.

2. Open and click tracking, and Primeforge

You: Do you need open and click tracking, or are delivery, bounce, and reply enough? yes i need this in my backend.... Which provider is "Prime Force", and what are its per-mailbox daily send limit and IMAP/OAuth access? for business mailbox use primeforge

Claude: I've updated the plan with open and click tracking and with Primeforge as your business mailbox provider.

Open and click tracking now has its own section. Each email gets a signed tracking pixel, and every link is rewritten to go through a redirect on your own tracking domains. Opens and clicks are saved as events and counts on each message, and they never overwrite its delivered, bounced or replied status. Automatic opens from Apple Mail and link clicks by security scanners are recorded but marked so they don't count in your reports. I also added the matching tables and fields to the schema, two tracking URLs and an engagement report to the API, and per-campaign `track_opens` / `track_clicks` switches.

Primeforge sells pre-warmed Google Workspace and Microsoft 365 mailboxes, so your backend connects to them like any Google or Microsoft account. That means the Gmail API, Microsoft Graph or IMAP. They cost about $3.50 to $4.50 per mailbox per month, so your 30 mailboxes come to about $105 to $135 a month plus domains.

I couldn't find a published daily send limit for Primeforge. The plan caps each mailbox at 30 a day (40 at most), which is about 900 a day from the business pool. At 5,000 a day, the normal pool carries the other 4,100. Please confirm the recommended limit with Primeforge before launch.

I also adjusted the capacity and warm-up tables to these numbers and ticked both open questions in the doc.

3. Capacity, limits, and replies

You: (pasted the capacity table) dont update the md file , explain properly here

Claude: This table answers one question: can your mailboxes send 2,000 to 5,000 emails a day without any single mailbox sending too much?

What each column means:

- **Mailboxes** is how many sender accounts are in each pool: 30 business mailboxes from Primeforge and 1,500 normal ones.

- **Daily cap per mailbox** is the most one mailbox may send in 24 hours once it is warmed up. The first number is the normal limit the system uses every day. The "hard ceiling" is an emergency maximum it never goes above.
- **Pool capacity per day** is mailboxes x cap, the most the whole pool could send if every mailbox hit its limit.
- **Load at 2,000/day** and **Load at 5,000/day** show how much the plan actually puts on each pool at your two target volumes, and how much that means per mailbox.

The business row: 30 mailboxes x 30 emails = 900 a day as the normal limit. At the 40 ceiling it is 30 x 40 = 1,200, so capacity is "900 to 1,200." The limit is low because these mailboxes send cold outreach. Cold-email practice is to keep each mailbox at roughly 20 to 40 a day; above that, Google and Microsoft start sending its mail to spam or blocking the account. At 2,000 a day, the business pool sends 600 (20 per mailbox); at 5,000 a day, it sends 900, the full 30 per mailbox.

The normal row: 1,500 mailboxes x 4 = 6,000 a day, and x 5 = 7,500 at the ceiling. The cap is very small because these are free accounts, which get locked fast if they send much. At 2,000 a day this pool sends 1,400 (about 1 per mailbox); at 5,000 a day it sends 4,100 (about 2.7 per mailbox).

The total row: 1,530 mailboxes together can send 6,900 to 8,700 a day. Your goal is at most 5,000, so you have spare room for mailboxes that are paused or locked. Even if 300 normal mailboxes were locked, the other 1,200 x 4 = 4,800 plus 900 business would still cover 5,000.

	Business	Normal	Total
At 2,000/day	600	1,400	2,000
At 5,000/day	900	4,100	5,000

In practice the split depends on the `mail_type` your MCP server sends with each request. Business-only volume above about 900 a day either waits for the next day or spills into the normal pool, depending on the `overflow` setting. The business cap of 30 is a safe estimate, not a published Primeforge number.

You: by 30 business mail i cand send 900-1200 email per day? what is load at 2000/day and 5000/day

Claude: Yes. 30 mailboxes x 30 emails = 900 per day (the normal, safe amount); 30 x 40 = 1,200 per day (the absolute maximum). "Load" means how many emails each pool actually sends on a given day. On a 2,000-email day the business pool sends 600 (20 each) and the normal pool 1,400 (about 1 each). On a 5,000-email day the business pool sends 900 (30 each, its full safe limit) and the normal pool 4,100 (2 or 3 each). Capacity is how much a pool can send; load is how much it does send that day.

You: i have any mailbox like 30 business mailbox and 1500 normal mailbox, how much i can send and pulled replies?

Claude: Sending: about 6,900 emails a day safely, 8,700 at the very most.

Pool	Mailboxes	Safe per mailbox	Safe total	Max per mailbox	Max total
Business (Primeforge)	30	30	900	40	1,200
Normal	1,500	4	6,000	5	7,500
Total per day	1,530		6,900		8,700

Replies: there is no real limit on how many you can pull. Receiving isn't capped like sending; the only question is how quickly the backend checks each inbox. The 30 business mailboxes keep a live connection open (Gmail push, Microsoft Graph, or IMAP IDLE), so replies arrive in seconds. The 1,500 normal mailboxes are checked in groups of about 100 per minute: every 15 minutes, and every 2 minutes for 48 hours after a send, so replies arrive within about 2 to 15 minutes.

Cold email usually gets roughly a 1% to 5% reply rate (a typical range, not a promise):

Daily sends	Replies (1% to 5%)	Bounce notices (2% to 5%)
2,000	20 to 100	40 to 100
5,000	50 to 250	100 to 250
6,900	about 70 to 345	about 140 to 345

You: that means per business mail i can send 30 mail by a single business account and send 4 mail by a single normal mail?

Claude: Yes. Each business mailbox sends up to 30 emails a day, and each normal mailbox up to 4. "Per day" means any 24 hours. These are safe limits; the system can go up to 40 and 5 if really needed, never above. They apply after warm-up:

Week	Per business mailbox	Per normal mailbox
1	15	1
2	25	2
3	30	3
4 onward	30	4

Emails are spread through the day: a business mailbox sends about every 20 minutes, a normal one every 2 to 4 hours.

4. The round-robin sending process

You: what about the sending process by sender? like for business mail a single mail can be send by 30 mail so dont send 30 mails by that mail at once, it should be round robin like send a single mail 1-30 sender then start the 2nd mail start with 1, and its same like the normal mail also

Claude: Yes, that is how the plan works. Each send uses the next mailbox in line, so no mailbox sends its 30 emails in a burst.

Business pool (30 mailboxes):

Round	Email numbers	Senders used
Round 1	emails 1 to 30	mailbox 1, 2, 3 … 30 (one email each)
Round 2	emails 31 to 60	mailbox 1, 2, 3 … 30 again
Round 3	emails 61 to 90	mailbox 1, 2, 3 … 30 again
…	…	…
Round 30	emails 871 to 900	mailbox 1 … 30 (each has now sent 30, so it stops)

Normal pool (1,500 mailboxes):

Round	Email numbers	Senders used
Round 1	emails 1 to 1,500	normal mailbox 1 … 1,500 (one each)

Round	Email numbers	Senders used
Round 2	emails 1,501 to 3,000	normal mailbox 1 … 1,500 again
Round 3	emails 3,001 to 4,100	normal mailbox 1 … 1,100

Timing over a 10-hour window: business sends about one email every 40 seconds from the pool, so each mailbox sends about every 20 minutes; normal sends about 7 a minute from the pool, so each mailbox sends about every 3 to 4 hours.

Four small differences from a strict 1, 2, 3 … 30 order:

1. **Skipping:** a paused or locked mailbox is skipped; the circle continues without it.
2. **Cooldown:** a mailbox can't send again until its rest time has passed (about 20 minutes for business, 2 to 4 hours for normal).
3. **Sticky sender for follow-ups:** a person emailed before always gets the same mailbox, so follow-ups stay in one thread and replies trace back cleanly.
4. **Small random shuffle:** a few minutes of random timing so the order isn't perfectly mechanical. For a strict order, keep a counter in Redis that goes 1 to 30 and back to 1, skipping full or paused mailboxes.

You: for this update the md file?

Claude: Updated the round-robin part of the Sender pool section in the plan doc with the step-by-step rotation, a timing table, and the rules that change the strict order.

You: db and api are in another md file,

Claude: Moved the database schema and API sections into their own doc, "Email Backend: Database Schema and API Reference", and linked it from the main plan. While moving them, the default daily cap became 30 and Primeforge was added as a provider.

5. Cheapest server and mailbox providers

You: now suggest me which provider is more cheaper and suitable for this backend server and mailbox

Claude: My pick: a Contabo Cloud VPS 20 for the server and Primeforge for the business mailboxes, about $125 to $165 a month in total. Your 1,500 normal mailboxes cost nothing extra.

The backend runs fine on one VPS with 4 to 6 vCPU and 8 to 12 GB RAM. Besides price, the provider must allow outgoing SMTP on ports 465 and 587, because normal Gmail and Outlook accounts send through SMTP.

Provider	Plan	Specs	Price per month	Mail ports	Verdict
Contabo	Cloud VPS 20	6 vCPU, 12 GB RAM, 100 GB NVMe	about €7 (~$8)	Open	Cheapest for the specs; best pick
Hostinger	KVM 2	2 vCPU, 8 GB RAM, 100 GB NVMe	$8.99 first term, then $14.99	Port 25 not blocked	Good backup choice
Hetzner	CPX32	4 vCPU, 8 GB RAM	about €36	25 and 465 blocked; 587 open	Good quality, now too expensive
DigitalOcean	—	—	—	25, 465 and 587 all blocked	Avoid

Sources: Contabo pricing (February 2026 figure; check current price), Hetzner price increase, Hetzner and DigitalOcean port policies, DigitalOcean SMTP block, Hostinger KVM 2 pricing, Hostinger port 25.

Run PostgreSQL and Redis on the same server; at this volume a managed database isn't needed. Send a nightly backup to cheap off-site storage.

Mailboxes: stay with Primeforge for business ($3.50 to $4.50 per mailbox a month); keep your 1,500 normal accounts; skip Amazon SES unless you later send to opted-in lists.

Item	Cost per month
Contabo Cloud VPS 20	about $8
Primeforge, 30 mailboxes	$105 to $135
Domains for business mailboxes (approximate, about 10 domains)	about $10 to $15
Off-site backups	about $1 to $5
Total	about $125 to $165

One thing to know whichever server you pick: 1,500 Gmail and Outlook accounts logging in from one data-center IP will trigger security checks on some of them. That's the main reason normal accounts get locked.

6. Tracking without a sending service, Google OAuth, Outlook and Yahoo

You: as normal mail dont use 3rd party provider how its direct delivared, replied clicked all these info?

Claude: Without a sending service, every signal comes from the SMTP reply when you send, the sender's own inbox, or your own tracking server.

Signal	Where it comes from	How the backend gets it	How fast	How reliable
Sent	Gmail's SMTP server answers 250 OK	The send worker reads the SMTP response	Instant	Exact
Delivered	No direct signal exists	Marked delivered if no bounce within 72 hours	72 hours	An assumption
Bounced	A bounce email lands in the sender's inbox	Inbox checker parses error code and original Message-ID	Usually minutes	Exact
Replied	The reply lands in the sender's inbox	Inbox checker matches it to the original email	2 to 15 minutes	Exact
Opened	Mail app loads your tracking image	Your server logs the request	Instant	Approximate
Clicked	Link goes through your server	Your server logs it, then redirects	Instant	Good after bot filtering
Spam complaint	Not available for free accounts	None	—	No signal

Bounces come from Gmail's "Mail Delivery Subsystem" (or Outlook's "Undeliverable" message); 5.x.x codes are hard bounces and 4.x.x soft. Replies carry your Message-ID in `In-Reply-To`, which matches them to the exact message. Opens and clicks never involve Gmail, so they work the same for every mailbox. Primeforge mailboxes work the same way, except their replies arrive in seconds because those inboxes stay connected.

You: with google oauth?

Claude: Google OAuth works: the backend uses the Gmail API with a token each mailbox owner approves once.

Signal	SMTP + IMAP (password)	Google OAuth (Gmail API)
Sent	SMTP answers 250 OK	`messages.send` returns message ID and thread ID
Delivered	Assumed after 72 hours	Same
Bounced	Inbox check every 2 to 15 minutes	Push notice within seconds
Replied	Inbox check every 2 to 15 minutes	Push notice within seconds; matched by thread ID
Opened / clicked	Your tracking server	Same

Replies arrive by push: `users.watch` sends inbox changes to a Pub/Sub topic, Google posts a notice to `/v1/webhooks/gmail`, and `history.list` fetches only new messages; the watch is renewed daily. OAuth needs a Google Cloud project, and reading scopes (`gmail.readonly` or `gmail.modify`) are restricted: app verification is required beyond 100 accounts and a periodic security assessment applies. Each of the 1,500 accounts must sign in once. For Primeforge Workspace mailboxes, domain-wide delegation lets one service account act for all 30 without per-mailbox sign-in, if Primeforge gives admin access. Suggested path: OAuth with delegation for business now; app passwords and IMAP for normal accounts, moving to OAuth after verification.

You: what happend if its a outlook or yahoo mail, i cant confirm you all my 1500 normal mail are gmail

Claude: It still works: the backend needs one connector per provider and detects the provider when you import each mailbox.

	Gmail	Outlook.com / Hotmail / Live	Yahoo / AOL	Others
Login method	App password, or Google OAuth	OAuth only (password login turned off)	App password with 2-step verification	Usually an app password
How to send	SMTP or Gmail API	Microsoft Graph, or SMTP with OAuth	SMTP	SMTP
Replies and bounces	Inbox check, or Gmail push	Graph push, or inbox check	Inbox check only	Inbox check only
Reply speed	2 to 15 min, or seconds with push	Seconds with push	2 to 15 min	2 to 15 min
Delivered	Assumed	Same	Same	Same
Opened / clicked	Your tracking server	Same	Same	Same

The bulk import sorts addresses by domain (gmail.com, outlook.com/hotmail.com/live.com/msn.com, yahoo.com/ymail.com/aol.com) and checks the MX record for other domains, then test-logs into each one and reports the mix. Outlook accounts need an app registered in Microsoft Entra and a one-time sign-in each. Yahoo accounts need app passwords and are polled. The cap of 4 a day is far below every provider's limit.

7. Third-party mailboxes, Nylas, and cheaper alternatives

You: maybe better to use 3rd party mailbox

Claude: For cold outreach at 2,000 to 5,000 a day, paid mailboxes on your own domains are a better base than 1,500 free accounts. Option A, more Primeforge-type mailboxes: one way to connect, fewer lockouts, better inbox placement, within provider rules, instant replies; "delivered" is still assumed.

Daily volume	Mailboxes needed (30/day each)	You have	Buy extra	Extra cost per month
2,000	about 67	30	about 37	about $130 to $165
5,000	about 167	30	about 137	about $480 to $615

Total at 5,000/day: about $645 to $835 a month including domains and server. Option B, a sending service like Amazon SES: real delivery and complaint reports for about $15 a month, but only for opted-in recipients; these services suspend accounts that send cold outreach. Recommendation: Option A for cold outreach, SES for opted-in contacts, and phase out the free accounts as the paid pool grows.

You: primeforge use for business mail what about normal mail like gmail, outlook, yahoo

Claude: There is no legitimate third-party provider for normal @gmail.com, @outlook.com or @yahoo.com mailboxes; those come only from Google, Microsoft and Yahoo, one per person. Marketplaces selling bulk or aged accounts break the providers' terms, get locked in waves, can be reclaimed by their original creator, and hurt deliverability. Realistic choices: (1) keep your own 1,500 accounts and connect them directly; (2) replace "normal" with Primeforge Google and Microsoft mailboxes on personal-style domains, which also enables ESP matching (source). Suggestion: start with option 1, grow option 2 in batches; 10 Primeforge mailboxes replace about 75 free accounts.

You: nylas?

Claude: Nylas works and removes most of the hard parts, but is expensive at 1,530 mailboxes. It is one API over Gmail, Outlook, Yahoo, iCloud and IMAP, with a bounce webhook, open, click and reply webhooks, and a pre-verified Google app that skips Google's verification. Each mailbox still signs in once, account locks don't go away, and "delivered" is still assumed. Pricing: $15 a month including 5 accounts, then $2 per account.

Pool through Nylas	Accounts	List price per month
30 Primeforge mailboxes	30	about $65
Primeforge pool for 5,000/day	about 167	about $340
30 business + 1,500 normal	1,530	about $3,065

Verdict: not worth it for 1,500 free accounts; a good fit for a paid pool of about 167 mailboxes. Check custom tracking domain support and the acceptable use policy first.

You: cheaper provider like nylas services

Claude: The cheapest for your case is EmailEngine, a self-hosted service with one flat price for unlimited mailboxes.

Service	How it's priced	30 mailboxes	~170 mailboxes	1,530 mailboxes
EmailEngine (self-hosted)	Flat yearly license	about $121/mo	about $121/mo	about $121/mo
Aurinko	$1.50 or $2 per account	$45 to $60/mo	$255 to $340/mo	$2,295 to $3,060/mo
Nylas	$15 base + $2 per account	about $65/mo	about $345/mo	about $3,065/mo
Unipile	€49 minimum, then about €5 per account	about €150/mo	about €850/mo	about €7,650/mo

Sources: EmailEngine pricing, Aurinko billing, Unipile pricing per EmailEngine's comparison.

EmailEngine connects through IMAP, SMTP, the Gmail API and Microsoft Graph and supports Gmail through app passwords without Google verification. You run it yourself (Node.js plus Redis, one connection per mailbox), Outlook still needs your own Microsoft app, and account locks don't go away. It would replace the plan's connector layer; everything else stays in your backend.

Item	Cost per month
EmailEngine license	about $121
Contabo Cloud VPS 30	about €14 (~$16)
Primeforge, 30 mailboxes	$105 to $135
Domains + backups	about $15 to $20
Total	about $255 to $290

Suggestion: EmailEngine if you keep the 1,500 free accounts; Aurinko becomes competitive if you move to a paid pool of about 170 mailboxes. Test with EmailEngine's free trial first.